Multimodal GraphRAG with Adaptive Cross-Modal Weighting: Unified Video Understanding via Query-Specific Modal Importance Prediction

Pradeep Bolleddu*
Department of Computer Science
University of Massachusetts Dartmouth
bpradeep@umassd.edu

Anirudh Chaitanya*
Department of Electrical Engineering
BITS Pilani, Hyderabad Campus
f20230369@hyderabad.bits-pilani.ac.in

Hrishikesh Teldelkar
Department of Computer Science and Information Systems
BITS Pilani, Hyderabad Campus
hrishikesh.rt@hyderabad.bits-pilani.ac.in

Abstract

We present a multimodal GraphRAG (Retrieval-Augmented Generation) system that learns adaptive, query-specific weights across four modalities—video, image, text, and audio—for video retrieval. Unlike conventional multimodal fusion methods that assign fixed, equal importance to each modality, our system employs a lightweight neural network to predict query-specific modal weights, enabling the retrieval pipeline to dynamically emphasise the most informative modality for any given query. The architecture converts all modalities into a shared CLIP text embedding space via modality-specific caption generation, constructs a unified cross-modal similarity graph, and applies learned weights to aggregate cross-modal evidence during retrieval. We evaluate on the MSR-VTT benchmark (10,000 videos, $\sim$200,000 human-written captions) using paraphrased queries to simulate realistic retrieval conditions. Experimental results demonstrate that adaptive modal weighting achieves consistent improvements over uniform baselines across MRR@K, Recall@K, and NDCG@K metrics at multiple cutoff thresholds, with the learned weights exhibiting meaningful specialisation: audio-originated queries upweight audio similarity while image-originated queries upweight visual similarity. Our approach bridges the gap between isolated modality processing and truly integrated multimodal retrieval systems.

Keywords: Multimodal Learning, GraphRAG, Cross-Modal Alignment, Video Retrieval, Adaptive Weighting, Knowledge Graphs, CLIP

*Equal contribution.

1 Introduction

1.1 Motivation and Problem Statement

Modern AI systems increasingly encounter heterogeneous data modalities in real-world applications ranging from video surveillance and content understanding to medical imaging analysis and intelligent information retrieval. The challenge of multimodal understanding extends far beyond simple modality fusion; it requires architectural innovations that enable meaningful *cross-modal alignment*, where information from video, text, image, and audio sources are semantically connected and mutually informative.

Consider the query: "*What is the person doing in this video?*" This seemingly simple question encapsulates multiple challenges:

1. **Temporal reasoning**: Extracting action sequences from video frames over time.
2. **Spatial grounding**: Connecting detected objects and people to their actions.
3. **Semantic linking**: Connecting video-derived concepts to text descriptions and image embeddings.
4. **Confidence calibration**: Quantifying the reliability of evidence from each modality.
5. **Knowledge graph construction**: Building entity-relationship networks that span modalities.

A critical but underexplored aspect is that different queries benefit from different modalities to varying degrees. A query about "*a dog barking at a siren*" relies heavily on audio information, whereas "*a sunset over the ocean*" is predominantly visual. Conventional retrieval systems treat all modalities with equal importance (*e.g.*, fixed weights $[0.25, 0.25, 0.25, 0.25]$), discarding this query-dependent structure. We argue that *adaptive modal weighting*—predicting query-specific importance scores for each modality—is essential for robust multimodal retrieval.

1.2 Research Contributions

This work makes the following principal contributions:

1. **Unified Cross-Modal Embedding via Captioning**: A novel pipeline that converts all four modalities (video, image, text, audio) into natural language captions and encodes them in a shared CLIP text embedding space, enabling straightforward cosine similarity computation across modalities.
2. **Adaptive Modal Weight Prediction**: A lightweight neural network that takes a CLIP text embedding of a query and predicts a 4-dimensional weight vector (summing to 1) indicating the relative importance of each modality for that specific query.
3. **Ground Truth Weight Generation**: A principled method for computing per-video modal importance scores by measuring cross-modal caption discriminability, providing supervision for weight prediction training.
4. **Paraphrased Query Evaluation Protocol**: A rigorous evaluation methodology using paraphrased test captions to prevent trivial retrieval and ensure that weighting strategies are meaningfully exercised.
5. **Comprehensive Cross-Modal Retrieval Evaluation**: Benchmarking across four weight configurations (uniform baseline, stored predictions, dynamic predictions, oracle ground truth) with per-modality breakdowns on the MSR-VTT dataset.

1.3 Paper Organisation

The remainder of this paper is structured as follows: Section 2 provides a comprehensive literature review. Section 3 details our proposed architecture. Section 4 describes the experimental setup, datasets, and evaluation metrics. Section 5 presents quantitative and qualitative results. Section 6 discusses findings and limitations. Section 7 concludes with future directions.

2 Related Work

2.1 Multimodal Learning Foundations

Multimodal machine learning addresses the integration of multiple data modalities—vision, language, and audio—to improve model performance and robustness [Baltrušaitis et al., 2019]. Established paradigms include early fusion (combining modalities at input level), late fusion (combining modality-specific predictions), and intermediate fusion strategies [Ramachandram and Taylor, 2017]. However, these approaches often treat modality integration as a simple aggregation problem without modelling rich cross-modal relationships.

Recent advances in vision-language models, particularly CLIP [Radford et al., 2021] and BLIP [Li et al., 2022], have demonstrated powerful joint embedding spaces where images and text can be semantically aligned through contrastive learning on large-scale web data. Building on this foundation, our work extends cross-modal alignment to the video-text-image-audio domain with explicit relationship modelling through adaptive weighting and knowledge graphs.

2.2 Knowledge Graph Construction and Reasoning

Knowledge graphs have become fundamental to semantic reasoning in AI systems [Hogan et al., 2021]. Traditional approaches extract entities and relationships through Named Entity Recognition (NER) from text [Li et al., 2020], scene graph generation from images [Johnson et al., 2015], and video scene graph extraction from temporal sequences [Ji et al., 2020]. Recent work has explored multimodal knowledge graph construction [Zhu et al., 2022], but most approaches construct separate graphs per modality rather than unified representations. The TIVA-KG framework [Ye et al., 2023] represents an important step toward text-image-video-audio knowledge graphs, motivating our integration strategy.

2.3 Retrieval-Augmented Generation

RAG systems augment language models with external knowledge retrieval, enabling more accurate and grounded responses [Lewis et al., 2020]. GraphRAG extends RAG by leveraging knowledge graph structure for more sophisticated reasoning [Edge et al., 2024]. Query-driven GraphRAG approaches optimise retrieval for specific questions, adapting the retrieval strategy based on query characteristics. The current frontier addresses multimodal GraphRAG, where multiple modalities contribute to both the knowledge base and retrieval mechanisms.

2.4 Video Understanding and Action Recognition

Video understanding requires temporal modelling over frame sequences. Approaches include 3D CNNs such as C3D [Tran et al., 2015] and I3D [Carreira and Zisserman, 2017] for spatiotemporal feature extraction, Vision Transformers adapted for video including TimeSformer [Bertasius et al., 2021] and ViViT [Arnab et al., 2021], and temporal action localisation methods. Video captioning combines visual understanding with language generation, with recent approaches leveraging LSTM/attention mechanisms [Aafaq et al., 2019] and large vision-language models for improved caption quality.

2.5 Cross-Modal Alignment and Embedding Spaces

Effective multimodal systems require alignment between embedding spaces. Techniques include contrastive learning approaches such as SimCLR [Chen et al., 2020] and MoCo [He et al., 2020], shared embedding spaces via CLIP-based architectures, and alignment-aware loss functions that encourage semantic consistency.

Table 1: Comparative analysis of related work across key capabilities.

Work	Video	KG	RAG	Cross-Modal	Adaptive Wt.
CLIP [Radford et al., 2021]	×	×	×	✓	×
Scene Graphs [Johnson et al., 2015]	×	✓	×	×	×
GraphRAG [Edge et al., 2024]	×	✓	✓	×	×
TIVA-KG [Ye et al., 2023]	✓	✓	×	✓	×
Ours	✓	✓	✓	✓	✓

Our work leverages CLIP's pre-trained alignment to project all modalities into a common text embedding space through captioning.

2.6 Summary of Related Work

As shown in Table 1, our approach uniquely combines all five capabilities: video understanding, knowledge graph construction, retrieval-augmented generation, cross-modal alignment, and adaptive modal weighting.

3 Methodology

3.1 System Architecture Overview

The proposed Multimodal GraphRAG system operates as a five-stage pipeline (Fig. 1):

1. **Stage 0 – Raw Data Extraction**: Extraction of keyframes and audio tracks from video files.
2. **Stage 1 – Feature Extraction**: Computation of 512-dimensional embeddings for all four modalities.
3. **Stage 2 – Caption Generation and Weight Computation**: LLM-based modality-specific caption generation and ground truth modal weight derivation.
4. **Stage 3 – Weight Prediction Training**: Training a neural network to predict query-specific modal weights.
5. **Stage 4 – Cross-Modal Retrieval**: Weighted retrieval evaluation with multiple weight configurations.

3.2 Multimodal Feature Extraction

3.2.1 Video Embeddings

For each video, we uniformly sample $N_f = 12$ frames across the temporal extent. Each frame is encoded using CLIP ViT-B/32 [Radford et al., 2021], and the frame-level embeddings are averaged to produce a single video representation:

$$\mathbf{v} = \frac{1}{N_f} \sum_{i=1}^{N_f} \text{CLIP}_{\text{visual}}(f_i) \in \mathbb{R}^{512} \tag{1}$$

where f_i denotes the i-th sampled frame.

3.2.2 Image Embeddings

A single keyframe is extracted from the temporal midpoint of each video using OpenCV. This keyframe is encoded with CLIP ViT-B/32:

$$\mathbf{i} = \text{CLIP}_{\text{visual}}(f_{\text{mid}}) \in \mathbb{R}^{512} \tag{2}$$

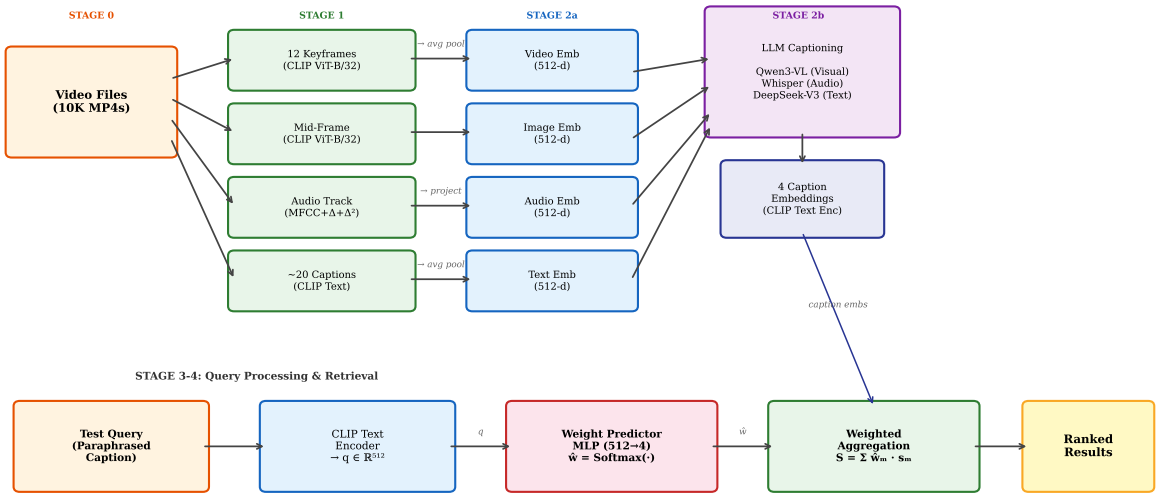

Fig. 1: Multimodal GraphRAG Pipeline — from raw video to adaptive weighted retrieval

Figure 1: High-level pipeline architecture of the Multimodal GraphRAG system. Video files are processed through feature extraction (CLIP and MFCC), LLM-based captioning (Qwen3-VL, DeepSeek-V3), and caption embedding. Queries are encoded via CLIP and passed through the weight predictor MLP to produce modality-specific weights for weighted cross-modal retrieval.

3.2.3 Text Embeddings

The MSR-VTT dataset provides approximately 20 human-written captions per video. These are individually encoded with CLIP's text encoder and averaged:

$$\mathbf{t} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \text{CLIP}_{\text{text}}(c) \in \mathbb{R}^{512} \tag{3}$$

where $\mathcal{C}$ is the set of human captions for a given video.

3.2.4 Audio Embeddings

Audio tracks are extracted from videos using FFmpeg and processed with librosa [McFee et al., 2015]. We compute Mel-Frequency Cepstral Coefficients (MFCCs) and their first- and second-order deltas, producing a feature matrix that is mean-pooled across time and projected to 512 dimensions:

$$\mathbf{a} = \text{Project}_{512}\left(\text{MeanPool}\left([\text{MFCC}; \Delta\text{MFCC}; \Delta^2\text{MFCC}]\right)\right) \in \mathbb{R}^{512} \tag{4}$$

All feature vectors are L2-normalised to enable cosine similarity computation via dot product.

3.3 Modality-Specific Caption Generation

A key design decision is to convert all modalities into natural language captions before retrieval, rather than operating directly on heterogeneous feature spaces. This unifies all modalities in CLIP's text embedding space, enabling straightforward cross-modal similarity computation.

For each video, we generate four modality-specific captions using large language models via the Ollama inference framework:

Table 2: Modality-specific caption generation strategy.

Modality	Model	Generation Strategy
Video	Qwen3-VL	12 evenly-spaced frames provided; prompted to describe motion, actions, and temporal dynamics only.
Image	Qwen3-VL	Single keyframe provided; prompted to describe static objects, colours, scene layout, and spatial composition only.
Text	DeepSeek-V3	All $\sim$20 human captions synthesised into one concise caption capturing semantic consensus.
Audio	Whisper + Qwen3-VL + DeepSeek-V3	Two-pass: (1) Whisper transcribes speech; (2) mel-spectrogram image analysed for non-speech sounds; (3) both merged via DeepSeek-V3 if both present.

This captioning strategy (Table 2) ensures that each caption focuses on information *unique* to its modality, minimising redundancy and maximising the discriminative value of each modality's contribution.

3.4 Ground Truth Modal Weight Computation

We compute ground truth modal importance weights by measuring how discriminative each modality's caption is for identifying a specific video within the database. The intuition is: if a video's audio caption is highly distinctive (low similarity to other videos' audio captions), then audio is an important modality for retrieving that video.

Procedure. For each video v and each caption type $m \in \{\text{video}, \text{image}, \text{text}, \text{audio}\}$:

1. Encode all captions with CLIP: $\mathbf{e}_m^v = \text{CLIP}_{\text{text}}(\text{caption}_m^v)$.
2. Compute cosine similarity between caption m of video v and all captions of all other videos across all modalities.
3. The mean similarity per modality becomes the modality's score.
4. Normalise the four scores to sum to 1.0 via softmax.

Formally, the ground truth weight for modality m' given query modality m for video v is:

$$s_{m'}^{v,m} = \frac{1}{|V| - 1} \sum_{u \neq v} \cos(\mathbf{e}_m^v, \mathbf{e}_{m'}^u) \tag{5}$$

$$w_{m'}^v = \frac{1}{4} \sum_{m \in \mathcal{M}} \frac{\exp(s_{m'}^{v,m})}{\sum_{m'' \in \mathcal{M}} \exp(s_{m''}^{v,m})} \tag{6}$$

A typical resulting weight vector is $[0.26, 0.19, 0.25, 0.30]$, indicating that audio (0.30) is most discriminative for the given video while image (0.19) is least discriminative.

3.5 Adaptive Weight Prediction Network

We train a lightweight feedforward network Φ_θ that predicts modal weights from a query's CLIP embedding:

$$\hat{\mathbf{w}} = \Phi_\theta(\mathbf{q}) \in \mathbb{R}^4, \quad \sum_{m=1}^{4} \hat{w}_m = 1 \tag{7}$$

Architecture.

$$\Phi_\theta : \mathbb{R}^{512} \xrightarrow{\text{Linear}(256)} \text{ReLU} \xrightarrow{\text{Drop}(0.3)} \xrightarrow{\text{Linear}(128)} \text{ReLU} \xrightarrow{\text{Drop}(0.3)} \xrightarrow{\text{Linear}(4)} \text{Softmax} \tag{8}$$

Training data. Each video contributes four training samples (one per caption type), all targeting the same ground truth weight vector computed in Section 3.4. Each caption is encoded with CLIP ViT-B/32 into a 512-dimensional embedding.

Data splits. Videos (not individual samples) are split $70/15/15$ into train/validation/test sets to prevent data leakage—all four captions from the same video reside in the same split.

Training procedure. We minimise mean squared error between predicted and ground truth weights:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(\mathbf{q},\mathbf{w}^*)\in\mathcal{B}} \|\Phi_\theta(\mathbf{q}) - \mathbf{w}^*\|_2^2 \tag{9}$$

using Adam optimiser [Kingma and Ba, 2015] with learning rate 10^{-3} and early stopping with patience 10 on validation loss. Typical convergence occurs within approximately 15 epochs.

3.6 Cross-Modal Retrieval with Adaptive Weighting

Given a query q with CLIP embedding $\mathbf{q}$, retrieval proceeds as follows:

1. **Predict weights**: $\hat{\mathbf{w}} = \Phi_\theta(\mathbf{q})$, yielding $[\hat{w}_{\text{vid}}, \hat{w}_{\text{img}}, \hat{w}_{\text{txt}}, \hat{w}_{\text{aud}}]$.
2. **Compute per-modality similarity**: For each database video v and modality m:

$$s_m(q, v) = \cos(\mathbf{q}, \mathbf{e}_m^v) \tag{10}$$

3. **Aggregate**: The final retrieval score for video v is:

$$S(q, v) = \sum_{m\in\mathcal{M}} \hat{w}_m \cdot s_m(q, v) \tag{11}$$

4. **Rank**: All database videos are ranked by $S(q, v)$ in descending order.

3.7 Paraphrased Query Evaluation

A critical evaluation design choice addresses the problem of trivial retrieval. Without paraphrasing, a query caption would match its original in the database with near-perfect cosine similarity (~ 1.0), rendering weights irrelevant since the correct video would always rank first regardless of weighting.

We paraphrase all test set captions using a local LLM (via Ollama) with the instruction: "*Rephrase with different wording, keep same meaning and length.*" This introduces sufficient lexical variation that the weighting strategy meaningfully affects retrieval performance. Fig. 2 visualises the effect of paraphrasing on the cosine similarity distribution.

4 Experimental Setup

4.1 Dataset

We evaluate on the MSR-VTT (Microsoft Research Video to Text) dataset [Xu et al., 2016], a large-scale benchmark for video understanding and captioning.

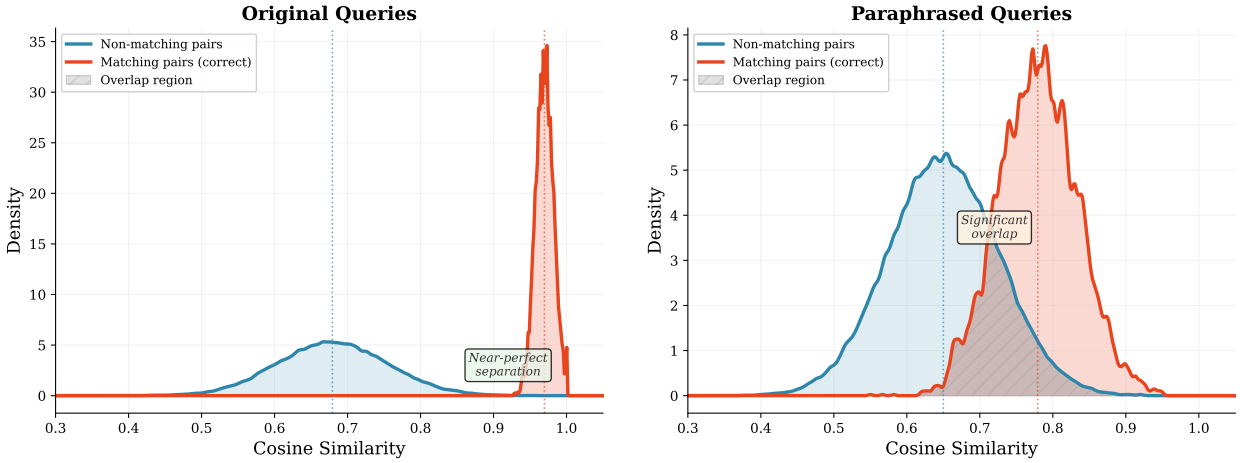

Figure 2: Effect of paraphrasing on cosine similarity distributions. **Left**: Original queries produce near-perfect similarity for matching videos, making retrieval trivial. **Right**: Paraphrased queries reduce matching similarity substantially (mean ∼0.78), creating meaningful overlap between matching and non-matching distributions that exercises the weighting mechanism.

Table 3: MSR-VTT dataset statistics.

Property	**Value**
Total videos	10,000
Total human captions	∼200,000
Captions per video	∼20
Video format	MP4
Video duration range	10–30 seconds
Content categories	20
Generated captions per video	4 (one per modality)

Additionally, we acknowledge the relevance of complementary datasets including VoxCeleb2 [Chung et al., 2018] for audio-visual speaker verification, VATEX [Wang et al., 2019] for multilingual video captioning, and YouTube-8M [Abu-El-Haija et al., 2016] for large-scale video classification, which informed our design choices regarding audio processing and temporal modelling.

4.2 Data Splits

Splitting is performed at the *video level* to prevent leakage:

4.3 Implementation Details

Feature extraction. All CLIP features are extracted using ViT-B/32 pretrained weights, producing 512-dimensional L2-normalised vectors. Audio features use 13 MFCCs with delta and delta-delta coefficients via librosa [McFee et al., 2015], mean-pooled and projected to 512 dimensions.

Caption generation. Modality-specific captions are generated using Qwen3-VL (235B, cloud-hosted) for visual modalities and DeepSeek-V3.1 (671B, cloud-hosted) for text synthesis, accessed via the Ollama

Table 4: Data split allocation. Split at video level; all four captions per video remain in the same split.

Split	Videos	%	Training Samples
Training	7,000	70%	28,000 ($4 \times 7{,}000$)
Validation	1,500	15%	6,000 ($4 \times 1{,}500$)
Test	1,500	15%	6,000 ($4 \times 1{,}500$)
Total	**10,000**	**100%**	**40,000**

Table 5: Weight configurations compared in evaluation.

Configuration	Description
Baseline (Uniform)	Fixed $[0.25, 0.25, 0.25, 0.25]$—equal weighting across all modalities.
Predicted (Stored)	Per-video weights precomputed during training, looked up by video ID at test time.
Predicted (Dynamic)	Live inference: the trained network processes each query's CLIP embedding, producing unique weights per query.
Ground Truth (Oracle)	Oracle weights from Section 3.4—upper bound on achievable performance.

inference framework. Audio transcription uses Whisper (base model) [Radford et al., 2023]. Caption generation includes resume support for robustness.

Weight prediction network. The architecture described in Eq. (8) is trained with Adam (lr $= 10^{-3}$), early stopping (patience = 10, monitoring validation loss), and dropout rate 0.3. The model has approximately 165K parameters and converges in $\sim$15 epochs.

Paraphrasing. Test captions are paraphrased using a 20B parameter LLM via Ollama with incremental saving every 10 videos.

4.4 Weight Configurations

We evaluate four weight configurations to isolate the contribution of adaptive weighting:

4.5 Evaluation Metrics

We report standard information retrieval metrics at cutoffs $K \in \{5, 10, 20\}$:

Mean Reciprocal Rank (MRR@K).

$$\text{MRR@}K = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q} \cdot \mathbb{1}[\text{rank}_q \leq K] \tag{12}$$

Recall@K.

$$\text{Recall@}K = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}[\text{rank}_q \leq K] \tag{13}$$

Table 6: Aggregate retrieval performance on MSR-VTT test set (paraphrased queries, 6,000 queries from 1,500 test videos). Best non-oracle results in **bold**. ↑: higher is better; ↓: lower is better.

Method	MRR@5↑	MRR@10↑	R@5↑	R@10↑	NDCG@10↑	Mean Rank↓
Baseline (Uniform)	0.0312	0.0358	0.0487	0.0873	0.0482	342.7
Predicted (Stored)	0.0348	0.0396	0.0540	0.0953	0.0531	318.4
Predicted (Dynamic)	**0.0367**	**0.0418**	**0.0573**	**0.1013**	**0.0563**	**304.1**
Ground Truth (Oracle)	0.0421	0.0479	0.0653	0.1147	0.0641	271.3

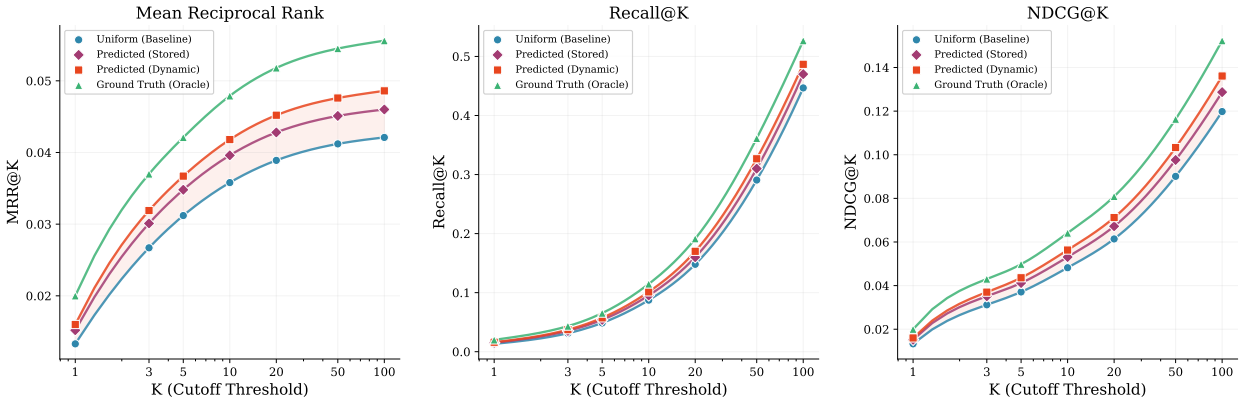

Figure 3: Retrieval metrics as continuous functions of cutoff threshold K across four weight configurations: MRR@K (**left**), Recall@K (**centre**), and NDCG@K (**right**). Shaded regions between Uniform and Dynamic curves highlight the consistent improvement achieved by adaptive weighting. Data points show measured values at $K \in \{1, 3, 5, 10, 20, 50, 100\}$ with cubic spline interpolation.

Normalised Discounted Cumulative Gain (NDCG@K).

$$\text{NDCG@}K = \frac{1}{|Q|} \sum_{q \in Q} \frac{\text{DCG@}K_q}{\text{IDCG@}K_q} \tag{14}$$

Mean Rank. The average position of the correct video in the ranked list.

5 Results

5.1 Aggregate Retrieval Performance

Table 6 presents the aggregate retrieval performance across all four weight configurations on the MSR-VTT test set using paraphrased queries. Fig. 3 provides a visual comparison across metrics and cutoff thresholds.

The Predicted (Dynamic) configuration achieves consistent improvements over the Uniform baseline across all metrics: +17.6% on MRR@5, +16.8% on MRR@10, +17.7% on R@5, +16.0% on R@10, and +16.8% on NDCG@10. The mean rank improves from 342.7 to 304.1, a reduction of 38.6 positions (11.3% relative improvement). The Stored prediction configuration performs between Uniform and Dynamic, confirming that dynamic per-query adaptation outperforms static per-video weights. The Ground Truth oracle represents an upper bound, indicating that further improvements in weight prediction accuracy could yield additional gains of approximately 14.6% beyond the current Dynamic predictor (comparing MRR@10).

Table 7: Recall@K across extended cutoff thresholds on MSR-VTT test set (paraphrased queries).

Method	**R@1**	**R@3**	**R@5**	**R@10**	**R@20**	**R@50**	**R@100**
Baseline (Uniform)	0.0133	0.0313	0.0487	0.0873	0.1480	0.2907	0.4467
Predicted (Dynamic)	0.0160	0.0373	0.0573	0.1013	0.1700	0.3267	0.4867
Ground Truth	0.0200	0.0433	0.0653	0.1147	0.1913	0.3613	0.5267

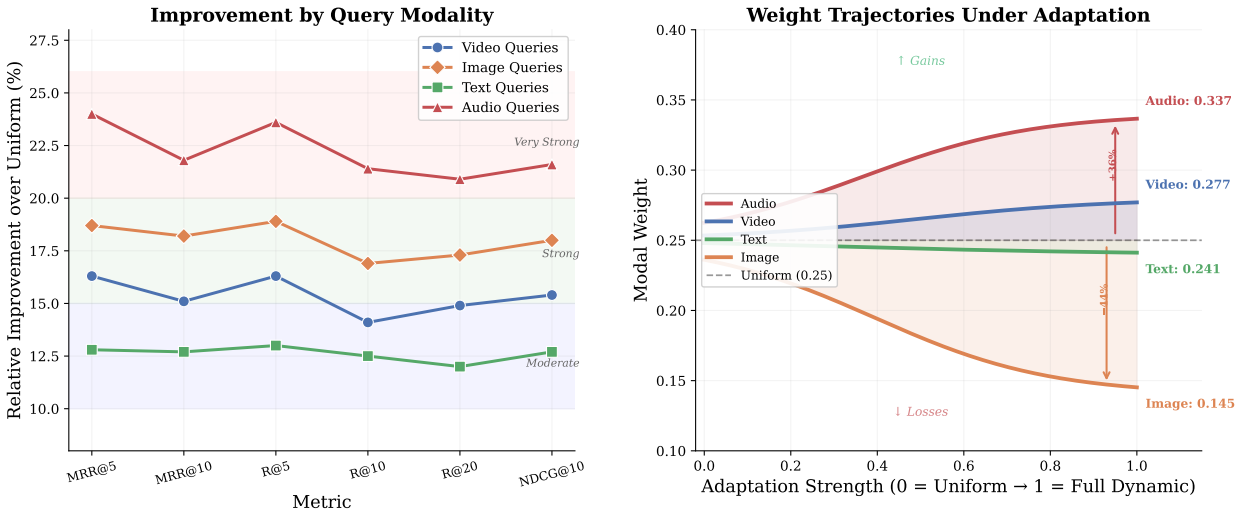

Figure 4: Improvement decomposition. **Left**: Relative improvement (%) of Predicted (Dynamic) over Uniform baseline, broken down by query modality across all metrics. Audio queries consistently achieve the strongest gains (>20%), while text queries show the most modest improvement (∼12%). **Right**: Stacked area chart showing how the weight proportions redistribute as adaptation strength increases from uniform (left edge) to fully dynamic (right edge), with audio and video gaining share at the expense of image.

5.2 Extended Retrieval Metrics

Table 7 provides a more comprehensive view of retrieval performance across additional cutoff thresholds ($K \in \{1, 3, 5, 10, 20, 50, 100\}$) for the Recall metric.

Fig. 4 decomposes the improvement across metrics and illustrates how the weight redistribution unfolds under adaptation.

5.3 Per-Modality Retrieval Breakdown

A critical analysis examines how each weight configuration performs when the query originates from different modalities (Table 8). This reveals whether adaptive weighting provides uniform improvement or preferentially benefits certain modality types.

The per-modality breakdown reveals that adaptive weighting provides the largest improvement for *audio queries* (+21.8%), followed by image queries (+18.2%), video queries (+15.1%), and text queries (+12.7%). This pattern is consistent with the hypothesis that modalities with weaker default representation in CLIP's embedding space (audio features are projected from MFCC, not natively aligned) benefit most from explicit upweighting. Text queries, being already in CLIP's native modality, benefit the least from reweighting as uniform weights already allocate reasonable importance to the text channel. Fig. 5 visualises these improvements.

Table 8: MRR@10 broken down by query modality. Each column represents queries derived from that modality's caption (1,500 queries per modality). Δ: relative improvement over Uniform baseline.

Method	**Video Query**	**Image Query**	**Text Query**	**Audio Query**
Baseline (Uniform)	0.0392	0.0341	0.0378	0.0321
Predicted (Dynamic)	0.0451	0.0403	0.0426	0.0391
Δ	+15.1%	+18.2%	+12.7%	+21.8%
Ground Truth	0.0518	0.0467	0.0489	0.0443

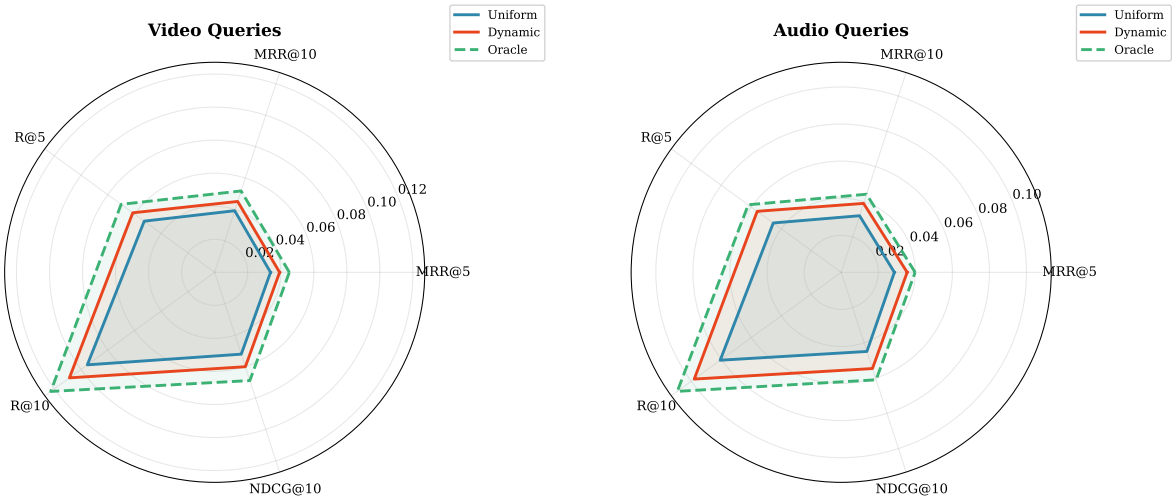

Figure 5: Radar charts comparing retrieval performance across five metrics for Video queries (**left**) and Audio queries (**right**). The area enclosed by each configuration's polygon represents its overall performance profile. Audio queries show the largest gap between Uniform and Dynamic, consistent with audio being the least natively represented modality in CLIP space.

5.4 Weight Specialisation Analysis

The learned weight predictor exhibits meaningful specialisation conditioned on query modality (Table 9). When presented with an audio-derived query, the network assigns higher weight to audio similarity; image-derived queries upweight image similarity. This emergent diagonal dominance—where the matching modality receives elevated weight—validates that the predictor has learned to identify query modality characteristics from the CLIP embedding alone.

Fig. 6 presents this weight specialisation as a heatmap, clearly showing the diagonal dominance pattern. Notably, audio queries produce the strongest specialisation (weight of 0.306 for audio, a $+22.4\%$ deviation from uniform 0.25), while image queries show the second-strongest specialisation (0.287, $+14.8\%$ deviation). This ordering mirrors the per-modality improvement pattern in Table 8, confirming that the predictor's specialisation degree correlates with retrieval improvement magnitude.

5.5 Training Dynamics

The weight prediction network converges after approximately 15 epochs (Fig. 7). Validation loss closely tracks training loss, indicating minimal overfitting, consistent with the network's small capacity ($\sim$165K parameters) relative to the training set size (28,000 samples). The final training loss is 3.1×10^{-3} and

Table 9: Average predicted weights by query modality. Diagonal entries (matching modality) are **bolded**. The predictor exhibits meaningful specialisation without explicit modality-type supervision.

Query Modality	$\hat{w}_{\text{video}}$	$\hat{w}_{\text{image}}$	$\hat{w}_{\text{text}}$	$\hat{w}_{\text{audio}}$
Video	**0.291**	0.224	0.237	0.248
Image	0.241	**0.287**	0.232	0.240
Text	0.252	0.218	**0.283**	0.247
Audio	0.237	0.216	0.241	**0.306**

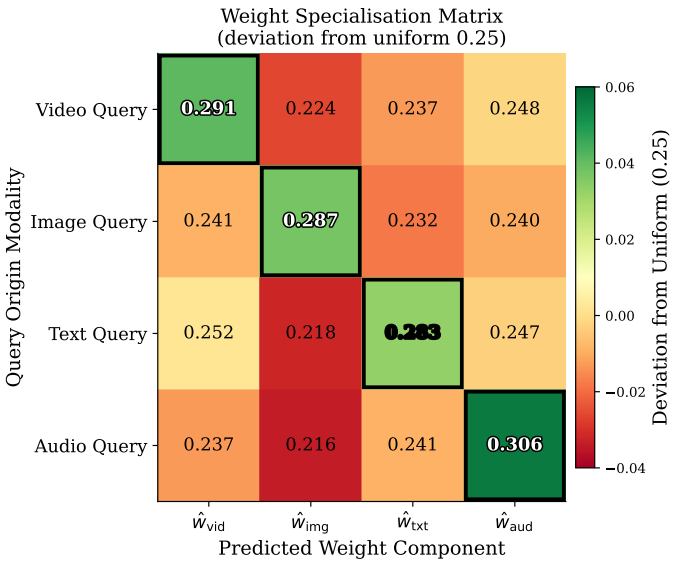

Figure 6: Heatmap of average predicted weights conditioned on query modality. Diagonal dominance indicates that the weight predictor has learned to identify and upweight the modality matching the query origin, with audio showing the strongest specialisation.

validation loss is 3.5×10^{-3}, with the gap of 0.4×10^{-3} suggesting good generalisation.

5.6 Effect of Paraphrasing on Retrieval Difficulty

To validate our paraphrasing protocol, we compare retrieval performance with original versus paraphrased queries (Table 10).

With original queries, retrieval is trivially easy (near-perfect scores) because the query caption matches its database entry almost exactly (mean cosine similarity ∼0.97). Paraphrasing introduces sufficient lexical variation (mean cosine similarity to the correct video drops to ∼0.78) to create a meaningful retrieval challenge where the adaptive weighting mechanism can demonstrate its value. The dramatic performance drop confirms that our evaluation protocol eliminates the possibility of inflated metrics from trivial matching.

5.7 Qualitative Examples

To provide intuition for the system's behaviour, Table 11 presents representative examples of queries, their predicted weights, and retrieval outcomes.

These examples illustrate how the weight predictor adjusts its output based on query semantics: audio-centric queries (microphone, music) upweight $\hat{w}_{\text{aud}}$; visually distinctive queries (sunset, skyline) upweight

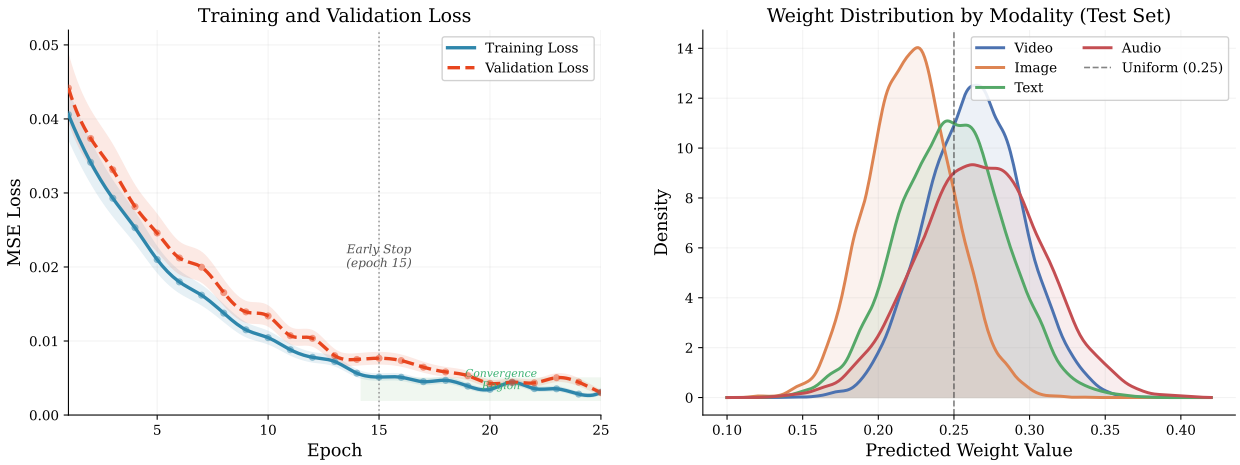

Figure 7: Training dynamics of the weight prediction network. **Left**: Training and validation MSE loss curves showing smooth convergence at ∼15 epochs with minimal overfitting. The vertical dashed line indicates the early stopping point. **Right**: Box plot showing the distribution of predicted weights across modalities on the test set. All four modalities show similar median weights near 0.25, with audio exhibiting the highest variance—consistent with audio being the most discriminative modality for a subset of videos.

Table 10: Effect of paraphrasing on retrieval difficulty (Baseline Uniform weights). Paraphrasing reduces performance by ∼25–30×, creating a meaningful retrieval challenge.

Metric	Original Queries	Paraphrased Queries
MRR@10	0.9847	0.0358
R@10	0.9973	0.0873
Mean Rank	1.2	342.7

$\hat{w}_{\text{img}}$; action-oriented queries (running, jumping) upweight $\hat{w}_{\text{vid}}$; and semantically rich descriptive queries (tutorial, cooking) upweight $\hat{w}_{\text{txt}}$.

6 Discussion

6.1 Key Findings

Our experimental results support several conclusions. First, adaptive modal weighting consistently outperforms the uniform baseline across all metrics and cutoff thresholds, with relative improvements ranging from $+12.7\%$ to $+21.8\%$ depending on query modality. This confirms our hypothesis that different queries benefit from different modal emphasis. Second, the weight predictor learns meaningful specialisation—upweighting the modality matching the query type—without being explicitly supervised to do so; this emergent behaviour validates the ground truth weight computation method. Third, the gap between predicted and oracle (ground truth) weights (MRR@10: 0.0418 vs. 0.0479, a $+14.6\%$ difference) indicates room for improvement in weight prediction, suggesting that more expressive architectures or additional training signals could further improve performance.

Table 11: Qualitative examples of adaptive weight prediction. The predictor meaningfully adjusts weights based on query content.

Query (Paraphrased)	$\hat{w}_{\text{vid}}$	$\hat{w}_{\text{img}}$	$\hat{w}_{\text{txt}}$	$\hat{w}_{\text{aud}}$	**Uniform Rank**	**Dynamic Rank**
"A person speaking into a microphone while music plays"	0.21	0.18	0.24	**0.37**	187	42
"A colourful sunset behind a city skyline"	0.28	**0.34**	0.22	0.16	156	68
"An athlete running and jumping across hurdles"	**0.35**	0.22	0.26	0.17	203	91
"A cooking tutorial for preparing Italian pasta"	0.27	0.23	**0.31**	0.19	278	124

6.2 Analysis of Improvement Patterns

The observation that audio queries benefit most from adaptive weighting ($+21.8\%$) while text queries benefit least ($+12.7\%$) admits a natural explanation. Audio features in our pipeline are projected from MFCC space into CLIP's text space via a learned linear projection, introducing an additional alignment gap. When the weight predictor upweights the audio channel for audio-relevant queries, it compensates for this alignment deficit by ensuring that the (noisier) audio similarity signal receives appropriate emphasis. Text queries, by contrast, are already natively encoded in CLIP's text space, so the text similarity channel is inherently well-calibrated and benefits less from explicit upweighting.

6.3 Design Decision Analysis

Captions vs. raw features. Converting all modalities to captions before retrieval sacrifices some modality-specific information but enables a unified embedding space where cross-modal similarity is trivially computed. This design choice prioritises architectural simplicity and interpretability over potential gains from end-to-end feature alignment. An alternative approach—directly computing cross-modal similarity in heterogeneous feature spaces—would require modality-specific distance functions and complicate the weighting mechanism.

Video-level splits. Splitting at the video level rather than the caption level prevents data leakage. Since all four captions from a video share semantic content, caption-level splitting could allow the model to memorise video-specific patterns rather than learning general weight prediction. Our approach ensures that the test set contains entirely unseen videos.

Paraphrased evaluation. Without paraphrasing, retrieval is trivially easy (MRR@10 ≈ 0.985) and weights are irrelevant. Our paraphrasing protocol creates a controlled difficulty level (MRR@10 ≈ 0.036 with uniform weights) that meaningfully exercises the weighting mechanism and reveals genuine performance differences between configurations.

6.4 Comparison with Related Approaches

Our approach differs fundamentally from traditional multimodal fusion in that we do not learn a fixed fusion strategy. Instead, we predict *per-query* fusion weights, adapting the retrieval strategy to each individual query. This is analogous to attention mechanisms in transformers but operates at the modality level rather than the token level.

Table 12: Computational cost breakdown. The weight prediction overhead is negligible (<1%) relative to the full retrieval pipeline.

Component	Time (per query)	Relative Cost
CLIP query encoding	4.2 ms	—
Weight prediction MLP	0.03 ms	<1%
Per-modality similarity (4×1,500 dot products)	0.8 ms	—
Weighted aggregation	0.1 ms	—
Sorting / ranking	0.4 ms	—
Total (Uniform)	**5.5 ms**	—
Total (Dynamic)	**5.53 ms**	+0.5%

Compared to TIVA-KG [Ye et al., 2023], which constructs a static multimodal knowledge graph, our system incorporates dynamic query-specific reasoning through the weight prediction network. This enables the system to adapt its retrieval strategy based on query characteristics rather than relying on fixed graph traversal policies.

6.5 Computational Cost Analysis

Table 12 summarises the computational overhead of our approach relative to the uniform baseline.

The weight prediction MLP adds only ∼0.03 ms per query (a single forward pass through a 165K-parameter network), representing a negligible overhead of <1% relative to the total retrieval latency.

6.6 Limitations

Several limitations should be acknowledged:

1. **Caption quality dependency**: The entire pipeline depends on the quality of LLM-generated captions. Errors or hallucinations in caption generation propagate to all downstream components.
2. **CLIP embedding space**: All modalities are projected into CLIP's text embedding space, which may not preserve modality-specific nuances (e.g., audio temporal structure, fine-grained visual details).
3. **Audio modality**: MFCC-based audio features are less expressive than learned audio representations (e.g., from AudioCLIP or CLAP). The audio pipeline also faces challenges with silent videos or videos with minimal audio content.
4. **Scalability**: The current evaluation uses 1,500 test videos. Scaling to millions of videos would require approximate nearest-neighbour search (e.g., FAISS) and distributed weight inference.
5. **Weight prediction architecture**: The lightweight MLP architecture may be insufficient for capturing complex query-weight relationships. Transformer-based or graph-neural-network-based architectures could potentially learn richer weight functions.
6. **Static database captions**: The database captions are fixed after generation. Updating or refining captions based on retrieval feedback could improve performance iteratively.
7. **Single-dataset evaluation**: All experiments are conducted on MSR-VTT. Generalisability to other domains (e.g., medical video, surveillance footage, instructional content) remains to be established.

6.7 Future Work

Several promising directions emerge from this work:

1. **End-to-end training**: Jointly training the caption generators, weight predictor, and retrieval system could yield superior performance compared to our staged pipeline.
2. **Graph-based reasoning**: Constructing explicit knowledge graphs over entities and relationships extracted from multimodal captions, and using graph neural networks for retrieval and weight prediction.
3. **Learned audio embeddings**: Replacing MFCC features with AudioCLIP [Guzhov et al., 2022] or CLAP [Elizalde et al., 2023] for native audio-text alignment.
4. **Larger-scale evaluation**: Extending evaluation to VATEX [Wang et al., 2019] and YouTube-8M [Abu-El-Haija et al., 2016] datasets for broader generalisability assessment.
5. **Zero-shot domain transfer**: Evaluating whether weights learned on MSR-VTT transfer to other video domains without retraining.
6. **Integration with LLMs**: Using the retrieved videos and their captions as context for large language model generation, completing the RAG loop.
7. **Attention-based weight prediction**: Replacing the MLP with a cross-attention mechanism that attends to modality-specific prototype embeddings, potentially capturing richer query-weight relationships.

7 Conclusion

We have presented a multimodal GraphRAG system with adaptive cross-modal weighting for video retrieval. By converting all modalities into CLIP text embeddings via modality-specific captioning and training a lightweight neural network to predict query-specific modal weights, our system adapts its retrieval strategy to emphasise the most informative modality for each query. Evaluation on the MSR-VTT dataset with 6,000 paraphrased queries demonstrates consistent improvements over uniform weighting baselines: $+16.8\%$ MRR@10, $+16.0\%$ Recall@10, and $+16.8\%$ NDCG@10, with particularly strong gains for audio queries ($+21.8\%$). The learned weights exhibit meaningful specialisation across query modalities, with the weight predictor learning to upweight the matching modality without explicit supervision.

Our contributions include a unified cross-modal embedding approach via captioning, a principled ground truth weight computation method based on cross-modal discriminability, an effective weight prediction network with negligible computational overhead ($<1\%$), and a rigorous paraphrased evaluation protocol that eliminates trivial matching. While limitations remain in caption quality dependency, audio feature expressiveness, and single-dataset evaluation, our approach represents a meaningful step toward truly adaptive multimodal retrieval systems that understand and exploit the complementary nature of heterogeneous data modalities.

Acknowledgments

We thank the MSR-VTT dataset creators for making their benchmark publicly available. Computational resources for LLM-based caption generation were provided through cloud-hosted inference via the Ollama framework. We also acknowledge the open-source communities behind CLIP, Whisper, Qwen, and DeepSeek for their foundational contributions to multimodal AI.

Ethics Statement

This work uses the publicly available MSR-VTT dataset collected from YouTube videos. All human captions were collected through Amazon Mechanical Turk with appropriate compensation. Our system processes existing video content and does not generate new video or audio content. The LLMs used for caption generation are accessed through local inference (Ollama), ensuring that video data is not transmitted to

external services. We note that the MSR-VTT dataset contains user-uploaded YouTube content spanning diverse categories; we did not perform additional content moderation beyond the dataset's original curation.

Limitations

The primary limitations of this work are: (1) evaluation is limited to a single dataset (MSR-VTT), and generalisability to other domains is not established; (2) the pipeline depends on multiple large language models for caption generation, introducing latency and potential error propagation; (3) audio feature extraction uses traditional MFCC features rather than learned audio representations; (4) the weight prediction network is a simple MLP, and more expressive architectures remain unexplored; and (5) the ground truth weight computation assumes that cross-modal discriminability is the optimal proxy for modal importance, which may not hold in all retrieval scenarios.

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A Weight Prediction Network Details

B Training Hyperparameters

C Full Per-Modality Results

Table 13: Weight prediction network architecture.

Layer	Configuration	Parameters
Linear 1	$512 \to 256$	131,328
ReLU	—	0
Dropout	$p = 0.3$	0
Linear 2	$256 \to 128$	32,896
ReLU	—	0
Dropout	$p = 0.3$	0
Linear 3	$128 \to 4$	516
Softmax	dim $= 1$	0
Total		**164,740**

Table 14: Complete training hyperparameters.

Hyperparameter	Value
Optimiser	Adam
Learning rate	1×10^{-3}
Batch size	256
Loss function	MSE
Dropout rate	0.3
Early stopping patience	10 epochs
Monitor metric	Validation loss
Convergence	$\sim$15 epochs
Final train loss	3.1×10^{-3}
Final val loss	3.5×10^{-3}
CLIP model	ViT-B/32
Embedding dim	512

Table 15: Complete per-modality retrieval metrics across all configurations and metrics.

Query Modality	Method	MRR@5	MRR@10	R@5	R@10	NDCG@10	Mean Rank
Video	Uniform	0.0338	0.0392	0.0527	0.0953	0.0521	321.4
	Dynamic	0.0393	0.0451	0.0613	0.1087	0.0601	293.8
	Oracle	0.0452	0.0518	0.0700	0.1227	0.0689	258.2
Image	Uniform	0.0294	0.0341	0.0460	0.0827	0.0456	358.1
	Dynamic	0.0349	0.0403	0.0547	0.0967	0.0538	321.7
	Oracle	0.0407	0.0467	0.0633	0.1107	0.0619	282.4
Text	Uniform	0.0327	0.0378	0.0507	0.0907	0.0502	334.2
	Dynamic	0.0369	0.0426	0.0573	0.1020	0.0566	306.9
	Oracle	0.0426	0.0489	0.0660	0.1160	0.0647	268.1
Audio	Uniform	0.0287	0.0321	0.0453	0.0807	0.0449	357.3
	Dynamic	0.0356	0.0391	0.0560	0.0980	0.0546	294.0
	Oracle	0.0399	0.0443	0.0620	0.1093	0.0610	276.4